

MODEL TWO STEP PROCESS:

1. Assignment to pools (τ, A_i).
2. Selection Score (S_i), then select top K.

Unified Selection Score Equation

$$S_i = A_i + \lambda(B - D)$$

S_i = selection score

A_i = individual performance metric

λ = environmental weight

environmental impact

$L_{net} = B - D$ ($B \approx$ benefits, $D \approx$ drawbacks)

Congestion Equation v1

$(B - D) = -L_C$ (congestion)

$$S_i = A_i - \lambda L_C$$

Knockout (sim): $\lambda = 0 \Rightarrow S_i = A_i$

(congestion out of selection)

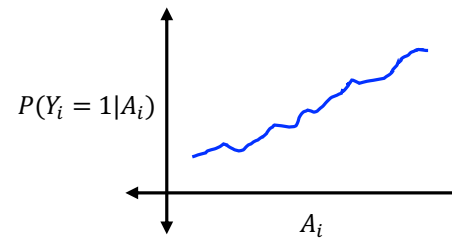
L_C (hard) $\rightarrow$ count with $A_j > \theta$

$\theta = \text{med}(A_i | Y_i = 1)$

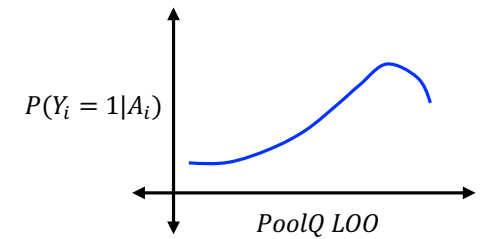
L_C (smooth, sim default) $\rightarrow$ LOO mean of $\sigma(\gamma(A_j - \theta))$ $j \neq i$

TALENT vs SELECTION

Naïve (empirical)



HERO (empirical)



Empirical (real panel)

Naïve: $P(Y_i = 1|A_i) \rightarrow \approx$ monotone

Hero: $P(Y_i = 1|PoolQ_LOO) \rightarrow$ tail dip